LAB ASSIGNMENT – 3
DATE – 02.02.2026

NAME : Soma Chatterjee
REGISTRATION NO. : AU/2023/0009439

ROLL NO. : UG/02/BTCSECSF/2023/013

SUBJECT : Unstructured Database Lab
SEC: E

Schema

{ roll_no, name, age, department, city, marks}

Initial Data

RollNo.	Name	Age	Department	City	Marks
1	Amit	20	CSE	Kolkata	78
2	Neha	19	IT	Delhi	65
3	Ravi	21	CSE	Mumbai	82
4	Pooja	22	ECE	Chennai	70
5	Karan	20	IT	Kolkata	55

Database Name: **collegeDB_'YourRollNo.'**

Collection Name: **students_' YourRollNo.'**

```
mongosh mongodb://127.0.0.

Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Soma Chatterjee>mongosh
Current Mongosh Log ID: 69d2ba8ab54ebfbe2a7c2906
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.7.0
Using MongoDB:          8.2.5
Using Mongosh:          2.7.0
mongosh 2.8.2 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-04-04T20:26:52.211+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
------

test> db.students.find()

test> use collegeDB_YourRollNo
switched to db collegeDB_YourRollNo
collegeDB_YourRollNo> db.createCollection("students_YourRollNo")
{ ok: 1 }
collegeDB_YourRollNo> db.students_YourRollNo.insertMany([
|   { roll_no: 1, name: "Amit", age: 20, department: "CSE", city: "Kolkata", marks: 78 },
|   { roll_no: 2, name: "Neha", age: 19, department: "IT", city: "Delhi", marks: 65 },
|   { roll_no: 3, name: "Ravi", age: 21, department: "CSE", city: "Mumbai", marks: 82 },
|   { roll_no: 4, name: "Pooja", age: 22, department: "ECE", city: "Chennai", marks: 70 },
|   { roll_no: 5, name: "Karan", age: 20, department: "IT", city: "Kolkata", marks: 55 }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69d2bb35b54ebfbe2a7c2907'),
    '1': ObjectId('69d2bb35b54ebfbe2a7c2908'),
    '2': ObjectId('69d2bb35b54ebfbe2a7c2909'),
    '3': ObjectId('69d2bb35b54ebfbe2a7c290a'),
    '4': ObjectId('69d2bb35b54ebfbe2a7c290b')
  }
}
```

Lab Questions:

Q1. Display all student records from the collection.

```
collegeDB> db.students_lab.find()
[
  {
    _id: ObjectId('698466d6ac63f4317e1e2621'),
    roll_no: '1',
    name: 'Amit',
    age: '20',
    department: 'CSE',
    city: 'Kolkata',
    marks: '78'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2622'),
    roll_no: '2',
    name: 'Neha',
    age: '19',
    department: 'IT',
    city: 'Delhi',
    marks: '65'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2623'),
    roll_no: '3',
    name: 'Ravi',
    age: '21',
    department: 'CSE',
    city: 'Mumbai',
    marks: '82'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2624'),
    roll_no: '4',
    name: 'Pooja',
    age: '22',
    department: 'ECE',
    city: 'Chennai',
    marks: '70'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2625'),
    roll_no: '5',
    name: 'Karan',
    age: '20',
    department: 'IT',
    city: 'Kolkata',
    marks: '55'
  }
]
```

Q2. Display all student records from the collection.

```
collegeDB> db.students_lab.find({}, { name: 1,department: 1,
... _id: 0})
[
  { name: 'Amit', department: 'CSE' },
  { name: 'Neha', department: 'IT' },
  { name: 'Ravi', department: 'CSE' },
  { name: 'Pooja', department: 'ECE' },
  { name: 'Karan', department: 'IT' }
]
```

Q3. Find students belonging to the CSE department.

```
collegeDB> db.students_lab.find({ department: "CSE" })
[
  {
    _id: ObjectId('698466d6ac63f4317e1e2621'),
    roll_no: '1',
    name: 'Amit',
    age: '20',
    department: 'CSE',
    city: 'Kolkata',
    marks: '78'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2623'),
    roll_no: '3',
    name: 'Ravi',
    age: '21',
    department: 'CSE',
    city: 'Mumbai',
    marks: '82'
  }
]
```

Q4. Find students whose age is greater than 20.

```
collegeDB> db.students_lab.find({ age: { $gt: 20 } })
```

Q5. Find students who scored less than 70 marks.

```
collegeDB> db.students_lab.find({ marks: { $lt: 70 } })
```

Q6. Display students belonging to Kolkata city.

```
collegeDB> db.students_lab.find({ city: "Kolkata" })
[
  {
    _id: ObjectId('698466d6ac63f4317e1e2621'),
    roll_no: '1',
    name: 'Amit',
    age: '20',
    department: 'CSE',
    city: 'Kolkata',
    marks: '78'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2625'),
    roll_no: '5',
    name: 'Karan',
    age: '20',
    department: 'IT',
    city: 'Kolkata',
    marks: '55'
  }
]
```

Q7. Sort students by age in ascending order.

```
collegeDB> db.students_lab.find().sort({ age: 1 })
[
  {
    _id: ObjectId('698466d6ac63f4317e1e2622'),
    roll_no: '2',
    name: 'Neha',
    age: '19',
    department: 'IT',
    city: 'Delhi',
    marks: '65'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2621'),
    roll_no: '1',
    name: 'Amit',
    age: '20',
    department: 'CSE',
    city: 'Kolkata',
    marks: '78'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2625'),
    roll_no: '5',
    name: 'Karan',
    age: '20',
    department: 'IT',
    city: 'Kolkata',
    marks: '55'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2623'),
    roll_no: '3',
    name: 'Ravi',
    age: '21',
    department: 'CSE',
    city: 'Mumbai',
    marks: '82'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2624'),
    roll_no: '4',
    name: 'Pooja',
    age: '22',
    department: 'ECE',
    city: 'Chennai',
    marks: '70'
  }
]
```

Q8. Sort students by marks in descending order.

```
collegeDB> db.students_lab.find().sort({ marks: -1 })
[
  {
    _id: ObjectId('698466d6ac63f4317e1e2623'),
    roll_no: '3',
    name: 'Ravi',
    age: '21',
    department: 'CSE',
    city: 'Mumbai',
    marks: '82'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2621'),
    roll_no: '1',
    name: 'Amit',
    age: '20',
    department: 'CSE',
    city: 'Kolkata',
    marks: '78'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2624'),
    roll_no: '4',
    name: 'Pooja',
    age: '22',
    department: 'ECE',
    city: 'Chennai',
    marks: '70'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2622'),
    roll_no: '2',
    name: 'Neha',
    age: '19',
    department: 'IT',
    city: 'Delhi',
    marks: '65'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2625'),
    roll_no: '5',
    name: 'Karan',
    age: '20',
    department: 'IT',
    city: 'Kolkata',
    marks: '55'
  }
]
```

Q9. Display only the first three student records.

```
collegeDB> db.students_lab.find().limit(3)
[
  {
    _id: ObjectId('698466d6ac63f4317e1e2621'),
    roll_no: '1',
    name: 'Amit',
    age: '20',
    department: 'CSE',
    city: 'Kolkata',
    marks: '78'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2622'),
    roll_no: '2',
    name: 'Neha',
    age: '19',
    department: 'IT',
    city: 'Delhi',
    marks: '65'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2623'),
    roll_no: '3',
    name: 'Ravi',
    age: '21',
    department: 'CSE',
    city: 'Mumbai',
    marks: '82'
  }
]
```

Q10. Update the marks of student Neha to 72.

```
collegeDB> db.students_lab.updateOne({ name: "Neha"}, { $set: {marks: "72" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Q11.Increase the marks of all students by 5.

```
collegeDB> db.students_lab.updateMany(
...     {},
...     { $inc: { marks: 5 } }
... )
...
```

Q12. Delete the students having rroll number 5.

```
collegeDB> db.students_lab.deleteOne({ roll_no: 5 })
{ acknowledged: true, deletedCount: 0 }
```

Q13. Find students whose age is between 20 and 22.

```
collegeDB> db.students_lab.find({
...     age: { $gte: 20, $lte: 22 }
... })
...
```

Q14. Display students belonging to either CSE or IT department.

```
collegeDB> db.students_lab.find({
...   department: { $in: ["CSE", "IT"] }
... })
...
[
  {
    _id: ObjectId('698466d6ac63f4317e1e2621'),
    roll_no: '1',
    name: 'Amit',
    age: '20',
    department: 'CSE',
    city: 'Kolkata',
    marks: '78'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2622'),
    roll_no: '2',
    name: 'Neha',
    age: '19',
    department: 'IT',
    city: 'Delhi',
    marks: '72'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2623'),
    roll_no: '3',
    name: 'Ravi',
    age: '21',
    department: 'CSE',
    city: 'Mumbai',
    marks: '82'
  },
  {
    _id: ObjectId('698466d6ac63f4317e1e2625'),
    roll_no: '5',
    name: 'Karan',
    age: '20',
    department: 'IT',
    city: 'Kolkata',
    marks: '55'
  }
```

localhost:27017 > collegeDB > students_lab

Documents 5 | Aggregations | Schema | Indexes 1 | Validation

Type a query: { field: 'value' } or Generate query

ADD DATA | UPDATE | DELETE | EXPORT DATA | EXPORT CODE

```
_id: ObjectId('698466d6ac63f4317e1e2621')
roll_no : "1"
name : "Amit"
age : "20"
department : "CSE"
city : "Kolkata"
marks : "78"
```

```
_id: ObjectId('698466d6ac63f4317e1e2622')
roll_no : "2"
name : "Neha"
age : "19"
department : "IT"
city : "Delhi"
marks : "72"
```

```
_id: ObjectId('698466d6ac63f4317e1e2623')
roll_no : "3"
name : "Ravi"
age : "21"
department : "CSE"
city : "Mumbai"
marks : "82"
```

llegeDB.students_lab

p

collegeDB | students_lab | mongosh: localhost:27017 | ai-detector

localhost:27017 > collegeDB > students_lab

Documents 5 | Aggregations | Schema | Indexes 1 | Validation

Type a query: { field: 'value' } or Generate query

Explain | Reset

ADD DATA | UPDATE | DELETE | EXPORT DATA | EXPORT CODE

25 | 1 – 5 of 5

```
marks : "72"

_id: ObjectId('698466d6ac63f4317e1e2623')
roll_no : "3"
name : "Ravi"
age : "21"
department : "CSE"
city : "Mumbai"
marks : "82"

_id: ObjectId('698466d6ac63f4317e1e2624')
roll_no : "4"
name : "Pooja"
age : "22"
department : "ECE"
city : "Chennai"
marks : "70"

_id: ObjectId('698466d6ac63f4317e1e2625')
roll_no : "5"
name : "Karan"
age : "20"
department : "IT"
city : "Kolkata"
marks : "55"
```